Dr. rer. nat. Martin Bies

Scientific Software Developer • Mathematical Physicist

- Kaiserslautern, Germany
- mbies87@gmail.com
- martinbies.github.io
- github.com/HereAround
- linkedin.com/in/martin-bies-000a1156
- German (native) • English (fluent) • French (B1) • Mandarin (beginner)

SUMMARY

I am a **scientific software developer and mathematical physicist** with a **Dr. rer. nat. in Physics** from Heidelberg University. Since 2022, I have been a core contributor to the open-source computer algebra system `OSCAR`, with particular responsibility for `FTheoryTools` and toric geometry. My work spans **software architecture**, algorithm and API design, testing and CI, performance optimisation, documentation, and technical project coordination. My research background spans **algebraic geometry and string theory** and has resulted in ten peer-reviewed publications and extensive international collaboration.

PROFESSIONAL EXPERIENCE

CURRENT, FROM 10/2022 (FT)

Mathematics Dept., RPTU Kaiserslautern-Landau, GER
Research Associate

Core contributor to the `OSCAR` open source computer algebra system (oscar-system.org), with major responsibility for the module `FTheoryTools` and toric geometry. My work includes algorithm and API design, data structures, testing and CI, performance optimisation, documentation, and technical coordination across the OSCAR software ecosystem. I also conceived and led a funded development project around `FTheoryTools`, including budgeting, recruitment and supervision of a research assistant, roadmap planning, and coordination with international collaborators.

09/2021 – 08/2022 (FT)

Dept. of Phys. & Astron., University of Pennsylvania, USA
Simons Postdoctoral Fellow

Continuation of *Simons Foundation* project, detailed below.

09/2020 – 08/2021 (FT)

Dept. of Mathematics, University of Pennsylvania, USA
Simons Postdoctoral Fellow

Work with M. Cvetič and R. Donagi on root bundles and the F-theory QSMs (funded by the *Simons Foundation*).

10/2019 – 09/2020 (FT)

Mathematical Institute, University of Oxford, UK
Long Term Visitor

Continuation of *Wiener-Anspach* project initiated at PTM, Brussels.

10/2018 – 09/2019 (FT)

PTM, Université Libre de Bruxelles, BE
Postdoctoral Researcher

M/F-Theory: Engineering of Super Conformal Field Theories (funded by the *Foundation Wiener-Anspach*).

02/2018 – 09/2018 (FT)

ITP, Heidelberg University, GER
Research Associate

AI-tools meet jumps in vector-like spectra (preparation of *Cluster of Excellence EXC 2181 STRUCTURES*).

EDUCATION

03/2014 – 02/2018 **Dr. rer. nat. in Physics (Grade: Magna Cum Laude)**
ADVISOR: PROF. T. WEIGAND (PHYSICS) & PROF. M. BARAKAT (MATHEMATICS)
Heidelberg University, GER

09/2012 – 02/2014 **Master of Physics (Grade: 1.0)**
ADVISOR: PROF. T. WEIGAND
Heidelberg University, GER

10/2010 – 06/2011 **ERASMUS exchange student**
Imperial College, London

10/2008 – 08/2012 **Bachelor of Physics (Grade: 1.1)**
ADVISOR: PROF. T. WEIGAND
Heidelberg University, GER

SCHOLARSHIPS AND AWARDS

04/2024 – 12/2024 **TU-Nachwuchsring**
Funded software-development project around `FTheoryTools`; conceived the project, prepared the proposal and budget, and recruited and supervised a research assistant. Project outcomes included software contributions, a peer-reviewed publication (2026), and an M.Sc. thesis (09/2025).

01/2010 – 02/2018 **Studienstiftung des deutschen Volkes**
2015: Awarded PhD scholarship.
2010: Awarded Bachelor and Master scholarship.

10/2010 – 06/2011 **ERASMUS scholarship**

SERVICES

2023–2025 **Expert Evaluator, Marie Skłodowska-Curie Actions**: evaluated 21 EU research proposals, typically requesting about EUR 250,000 in funding; prepared structured assessments and participated in and led consensus discussions.

SINCE 2024 Reviewer for *Journal of High Energy Physics* and Mathematical Reviews/MathSciNet.

2016–2025 **Admission Board Member, Studienstiftung des deutschen Volkes** (selected years): Conducted approximately 60 structured interviews with undergraduate students and contributed to committee decisions on the award of merit-based scholarships.

07/2024 Co-organizer of a session at *ICMS 2024*, together with M. Zach and L. Kastner.

SINCE 2021 10+ letters of recommendation for students.

PUBLICATIONS

ORCID	0000-0002-9609-1693
SCOPUS	57197835420
H-INDEX	5 (Based on peer-reviewed works, only.)
TOTAL PUBLICATIONS	17
PEER REVIEWED/ACCEPTED	10
OUTREACH	3
UNPUBLISHED	1
THESIS	3
JOURNALS	Communications In Mathematical Physics (1)
	Fortschritte der Physik (1)
	Journal of Algebra and Its Applications (1)
	Journal of High Energy Physics (5)
	Physical Review D (1)
	Proceedings of Symposia in Pure Mathematics (AMS) (1)

Peer Reviewed Publications

1 **M. Bies**, Miķelis E. Miķelsons, Andrew P. Turner, *FTheoryTools: Advancing Computational Capabilities for F-Theory Research*, Fortschritte der Physik, Jan. 2026, DOI: 10.1002/prop.70065.

2 **M. Bies**, M. Cvetič, R. Donagi, M. Ong, *Improved Statistics for F-theory Standard Models*, *Communications in Mathematical Physics*, Nov. 2024, DOI: 10.1007/s00220-024-05148-7.

3 **M. Bies**, *Root bundles: Applications to F-theory Standard Models*, in *String-Math 2022*, R. Donagi, A. Langer, P. Sułkowski, and K. Wendland, eds., Proceedings of Symposia in Pure Mathematics, vol. 107, American Mathematical Society, 2024, pp. 17–43. ISBN: 978-1-4704-7240-5. DOI: 10.1090/pspum/107. (A preprint is available at arXiv:2303.08144.)

4 **M. Bies**, M. Cvetič, R. Donagi, M. Ong, *Brill-Noether-general Limit Root Bundles: Absence of vector-like Exotics in F-theory Standard Models*, Journal of High Energy Physics, Nov. 2022, DOI: 10.1007/JHEP11(2022)004.

5 **M. Bies**, M. Cvetič, M. Liu, *Statistics of Root Bundles Relevant for Exact Matter Spectra of F-theory MSSMs*, Physical Review D, Sept. 2021, DOI: 10.1103/PhysRevD.104.L061903.

6 **M. Bies**, M. Cvetič, R. Donagi, M. Liu, M. Ong, *Root Bundles and Towards Exact Matter Spectra of F-theory MSSMs*, Journal of High Energy Physics, Sept. 2021, DOI: 10.1007/JHEP09(2021)076

7 **M. Bies**, S. Posur, *Tensor Products of Finitely Presented Functors*, Journal of Algebra and Its Applications, July. 2021, DOI: 10.1142/s0219498822501869.

8 **M. Bies**, M. Cvetič, R. Donagi, L. Ling, M. Liu, F. Ruehle, *Machine Learning and Algebraic Approaches towards Complete Matter Spectra in 4d F-theory*, Journal of High Energy Physics, Jan. 2021, DOI: 10.1007/JHEP01(2021)196.

9 **M. Bies**, C. Mayrhofer, T. Weigand, *Algebraic Cycles and Local Anomalies in F-theory*, Journal of High Energy Physics, Nov. 2017, DOI: 10.1007/jhep11(2017)100.

10 **M. Bies**, C. Mayrhofer, T. Weigand, *Gauge Backgrounds and Zero-Mode Counting in F-theory*, Journal of High Energy Physics, Nov. 2017, DOI: 10.1007/jhep11(2017)081.

Outreach

11 **M. Bies**, A. P. Turner, *F-Theory Applications*, in *The Computer Algebra System* `OSCAR`*: Algorithms and Examples*, W. Decker, C. Eder, C. Fieker, M. Horn, and M. Joswig, eds., Algorithms and Computation in Mathematics, vol. 32, Springer, 1st ed., 2025, pp. 453–475. ISSN: 1431-1550.

12 **M. Bies**, L. Kastner, *Toric Geometry*, in *The Computer Algebra System* `OSCAR`*: Algorithms and Examples*, W. Decker, C. Eder, C. Fieker, M. Horn, and M. Joswig, eds., Algorithms and Computation in Mathematics, vol. 32, Springer, 1st ed., 2025, pp. 193–213. ISSN: 1431-1550.

13 **M. Bies**, L. Kastner, *Toric Geometry in* `OSCAR`, ComputerAlgebraRundbrief 72 (03/2023), 20-25, Mar. 2023, Preprint: https://arxiv.org/abs/2303.08110.

Unpublished Works

14 **M. Bies**, C. Mayrhofer, C. Pehle, T. Weigand, *Chow Groups, Deligne Cohomology and Massless Matter in F-theory*, Feb. 2014, https://arxiv.org/abs/1402.5144.

Thesis

15 **M. Bies**, *Cohomologies of Coherent Sheaves and Massless Spectra in F-theory*, PhD thesis, Heidelberg University, Feb. 2018, Heidelberg University Library, available at DOI: 10.11588/HEIDOK.00024045.

16 **M. Bies**, *Cohomologies of holomorphic line bundles in smooth and compact normal toric varieties*, M.Sc. thesis, Heidelberg University, February 2014, available on author's academic homepage.

17 **M. Bies**, *Intersecting D6-brane models on* $T^2 \times T^2 \times T^2 / (\sigma \times \Omega)$ *and* $T^2 \times T^2 \times T^2 / (\mathbb{Z}_2 \times \mathbb{Z}_2 \times \sigma \times \Omega)$ *orientifolds*, B.Sc. thesis, Heidelberg University, August 2012, available on author's academic homepage.

TEACHING RECORD

Autonomous Instruction of Lecture Courses

Period	Title	University	Students	Weekly Teaching	Evaluation
04/2024 – 07/2024	*Introduction to Topology*	*RPTU KL-LD, GER*	5	1×1.5 hours	Insufficient data due to sharp drop in RPTU KL-LD student numbers.
01/2022 – 05/2022	*Computational Linear Algebra*	*University of Pennsylvania, USA*	29	2×1.5 hours	2.12
01/2021 – 05/2021	*Computational Linear Algebra*	*University of Pennsylvania, USA*	57	2×1.5 hours	2.04

Scale: Poor (0), Fair (1), Good (2), Very good (3), Excellent (4).

Senior Teaching Assistant

Period	Title	University	Students	Weekly Teaching
10/2023 – 03/2024	*Algebraic Geometry*	*RPTU KL-LD, GER*	6	1×1.5 hours
04/2018 – 10/2018	*Methods of Math. Phys.*	*Heidelberg University, GER*	51	1×1.5 hours
04/2016 – 09/2016	*General Relativity*	*Heidelberg University, GER*	132	1×1.5 hours

Teaching Assistant

Period	Title	University	Weekly Teaching
10/2016 – 03/2017	*Theoretical Physics I*	*Heidelberg University, GER*	1×1.5 hours
04/2015 – 09/2015	*Theoretical Physics IV*	*Heidelberg University, GER*	1×1.5 hours
10/2014 – 03/2015	*Quantum Field Theory*	*Heidelberg University, GER*	1×1.5 hours
10/2013 – 03/2014	*Theoretical Physics III*	*Heidelberg University, GER*	1×1.5 hours
04/2013 – 09/2013	*Theoretical Physics II*	*Heidelberg University, GER*	1×1.5 hours
10/2012 – 03/2013	*Theoretical Physics I*	*Heidelberg University, GER*	1×1.5 hours

TALKS, POSTERS, CONFERENCES

Organizer of Conference (1)

07/2024 Session at ICMS 2024 (Durham, UK)

Invited Talks (9)

09/2025 *Annual meeting 2025 of SFB-TRR 195* (Tuebingen, GER)
Title: *Computational Frontiers in Singular Elliptic Fibrations and F-Theory Model Building*

07/2023 *Third Annual Meeting 2023 of SFB-TRR 195* (Saarbruecken, GER)
Title: *F-Theory: Exemplifying `OSCAR`'s Pursuit for Multidisciplinary Excellence*

05/2023 *Oberseminar algebraische Geometrie* (Saarbruecken, GER)
Title: *F-Theory and Singular Elliptic Fibrations*

10/2020 Philadelphia, USA
Title: *Machine Learning and Algebraic Approaches towards Complete Matter Spectra in 4d F-theory*

06/2020 *Summer Series on String Phenomenology (Virtual)*
Title: *On Stratification Diagrams, Algorithmic Spectrum Estimates and Vector-Like Pairs in F-theory*

12/2019 Philadelphia, USA
Title: *From F-theory Standard Models to Freyd Categories and back*

10/2018 Brussels, BE
Title: *Counting Massless Matter in F-theory with* `CAP`

08/2018 *`CAP_days` 2018* (Siegen, GER)
Title: `CAP`, *Machine Learning and String Theory*

07/2014 Aachen, GER
Title: *The Standard Model from String Theory*

Other Talks at Conferences, Workshops etc. (15)

06/2024 *StringPheno 2024* (Pardova, IT)
Title: *Efficiency in F-Theory: `FTheoryTools`*

07/2023 *StringMath 2023* (Melbourne, AU)
Title: *Root bundles: Applications to F-theory Standard Models*

07/2023 *StringPheno 2023* (Daejeon, KR)
Title: *Root bundles: Applications to F-theory Standard Models*

05/2023 *Computeralgebra Tagung 2023* (Hannover, GER)
Title: *F-Theory Tools: String theory Applications of `OSCAR`*

07/2022 *String Math 2022* (Warsaw, PL)
Title: *Towards F-theory MSSMs*

07/2022 *String Pheno 2022* (Liverpool, UK)
Title: *Towards F-theory MSSMs*

09/2021 *Summer Series on String Phenomenology (virtual meeting)*
Title: *Root Bundles and Towards Exact Matter Spectra of F-theory MSSMs*

12/2020 *String Data 2020* (virtual conference)
Title: *Vector-like spectra in F-theory* (joined with M. Liu)

08/2019 *Gap Singular Meeting and School* (Lambrecht, GER)

Title: *Monoidal Structures in Freyd Categories*

05/2018 Seminar on *Holography and Large-N duality* (Heidelberg, GER)
Title: *Conformal Invariants; Fefferman–Graham Expansion; Graham–Lee Theorem* (with M. Zikidis)

07/2017 *String Pheno 2017* (Virginia, USA)
Title: *Zero Mode Counting in F-Theory via* `CAP`

08/2014 *`GAP` Days* (Aachen, GER)
Title: *String Theory, Sheaf Cohomology and the* `homalg` *Package*

05/2014 Seminar Series *What is?* (Heidelberg, GER)
Title: *What is a Fermion/Boson (in Quantum Mechanics)?*

02/2014 Heidelberg, GER
Title: *Cohomology of Holomorphic Pullback Line Bundles on Smooth, Compact Normal Toric Varieties*

05/2012 Heidelberg, GER
Title: *Intersecting D6-Brane Models*

Posters at Conferences, Workshops etc. (2)

07/2023 *StringMath 2023* (Melbourne, AU)
Title: `FTheoryTools` – *A Computer Tool for Singular Elliptic Fibrations*

09/2019 *Strings and Geometry* (Oxford, UK)
Title: *Tensor Products of Finitely Presented Functors*

Conferences and Workshops attended without Talk or Poster Contribution (17)

04/2025 *OSCAR Coding Sprint* (Kaiserslautern, GER)

03/2025 *Retreat of the SFB-TRR 195* (Bad Münster am Stein, GER)

12/2024 *LEAN meets MaRDI and OSCAR* (Berlin, GER)

11/2024 *MaRDI Annual Meeting 2024* (Kaiserslautern, ITWM, GER)

09/2024 *OSCAR Workshop* (Kaiserslautern, GER)

02/2023 *OSCAR developer meeting* (Kaiserslautern, GER)

07/2022 *Strings 2022* (Vienna, AT)

06/2022 Simons Collab.: *Geometry, Topology and Singular Special Holonomy Spaces* (Freiburg, GER)

11/2021 Simons Collab. (Homological Mirror Symmetry) *Annual Meeting* (New York, USA)

09/2021 Simons Collab.: *Progress and Open Problems* (Stony Brook, USA)

09/2021 Simons Collab. (Special Holonomy in Geometry, Analysis, Phys.) *Annual Meeting* (New York, USA)

07/2021 *String Pheno 2021* (virtual conference)

06/2021 *Strings 2021* (virtual conference)

06/2021 *String Math 2021* (virtual conference)

06/2020 *String Pheno 2020* (virtual conference)

07/2019 *Strings 2019* (Brussels, BE)

03/2018 *String Data 2018* (Munich, GER)

12/2015 *String Math 2015* (Sanya, CN)

09/2015 *Third `GAP` Days* (Trondheim, NO)

03/2015	*Second GAP Days* (Aachen, GER)
02/2015	*Physics and Geometry of F-Theory* (Munich, GER)
12/2014	*Homological Perturbation Theory* (Galway, IE)
02/2014	*Geometry and Physics of String Compactifications* (Heidelberg, GER)

OTHER TRAININGS

2024 – 2025	Mandarin Chinese courses, introductory level (RPTU Kaiserslautern).
03/2024	Moderation of meetings and project discussions (Kaiserslautern, GER). Offered by: *TU Nachwuchsring*
05/2018	Kontaktseminar – Schwerpunkt Banken und Beratung (Bonn, GER) Offered by: *Studienstiftung des deutschen Volkes*
05/2018	Physiker im Beruf (Bad Honnef, GER) Offered by: *Deutsche Physikalische Gesellschaft (DPG)*
05/2017	Training for admission board members (Frankfurt, GER) Offered by: *Studienstiftung des deutschen Volkes*

REFERENCES

Prof. Dr. Mirjam Cvetič
POSITION Fay R. and Eugene L. Langberg Professor
EMPLOYER Department of Physics and Astronomy
University of Pennsylvania, USA
EMAIL cvetic@physics.upenn.edu
PHONE +1 (215) 898 8153

Prof. Dr. Ron Donagi
POSITION Thomas A. Scott Professor of Mathematics
EMPLOYER Department of Mathematics
University of Pennsylvania, USA
EMAIL donagi@math.upenn.edu
PHONE +1 (215) 898 8465

Prof. Dr. Max Horn
POSITION Professor
EMPLOYER Department of Mathematics
RPTU Kaiserslautern-Landau, GER
EMAIL mhorn@rptu.de
PHONE +49 631 205 2730